Tomorrow's class — NC443 (jointly).

Recurrence Relations.

Writing a_n $(n \geq 0)$ as a dependence on some prior terms of the sequence $(a_{n-1}, a_{n-2}, \ldots)$.

Recurrence relation — Difference equations.

E.g: $a_{n+1} = 3a_n$, $n \geq 0$, $a_0 = 5$.

$n \geq 0$ → constraint.

$a_0 = 5$ → base condition / initial condition.

a_{n+1} → Next term is dependent on the current term / present term ONLY.

ORDER → First order recurrence relation

$a_{n+1} = 3a_n \quad n \geq 0, \; a_0 = 5$

(1) First order recurrence relation

(2) Linear

(3) Homogeneous.

$a_1 = 3a_0 = 3.5$

$a_2 = 3a_1 = 3(3a_0) = 3^2(5)$

$a_3 = 3a_2 = 3(3^2(5)) = 3^3(5)$

$\vdots \qquad \vdots$

$a_n = 5(3^n), \quad n \geq 0.$

Conclusion:

The general solution of a recurrence relation of the form

$$\boxed{a_{n+1} = Aa_n}$$

where A is a constant

$a_0 = D, \quad n \geq 0$

Soln: $a_n = DA^n.$

Solve: (ii) $a_{n+1}^2 = 5a_n^2$ where $a_n > 0$, $n \geq 0$, $a_0 = 2$.

(i) $a_n = 7a_{n-1}$, $n \geq 1$, $a_2 = 98$.

$a_n = 2(7^n)$.

$a_n = a_0(7^n)$. / a_2 is given. find a_0 & put it back in the sol^n expression.

$a_2 = a_0 \times 7^2$

$98 = 49a_0 \Rightarrow a_0 = 2$.

Sol^n: $a_n = 2(7^n)$.

Hint: Assume $b_n = a_n^2$

$b_{n+1} = 5b_n$

$b_0 = a_0^2 = 4$

$b_n = 4 \cdot 5^n$

$a_n = 2(\sqrt{5})^n$.

Homogeneous vs. Non-homogeneous.

Homogeneous	Non-homogeneous
$a_{n+1} + \dots + ca_n = f(n)$	$a_{n+1} + \dots + ca_n = f(n)$.
	c is constant $f(n)$ is a function where n is a nonnegative integer.
$f(n) = 0$	$f(n) \neq 0$

A is an array of n integers

```
Begin
  For i = 1 to n-1 do
    For j = n downto i+1 do
      if A[j] < A[j-1] then
        Begin
          temp = A[j-1]
          A[j-1] = A[j]
          A[j] = temp
        End
End
```

→ bubble sort.

↳ Write the time complexity as a recurrence relation.
↳ in terms of no. of comparisons

→ Q1. How many comparisons to bubble the smallest element to the beginning of the list. → $n-1$ → smallest element coming to its correct position.

Remaining ~~subsit~~ sublist of $n-1$ elements

$f(n)$

$a_n = a_{n-1} + (n-1)$ ✓

a_n : no. of comparisons. a_{n-1} : no. of comparisons for the subsit

$a_1 = 0$

$a_2 = a_1 + (2-1) = 1$

$a_3 = a_2 + (3-1) = 1+2$

⋮

$a_n = 1 + 2 + \cdots + (n-1)$

$= (n^2 - n)/2 \rightarrow O(n^2)$.

Second order recurrence relations (homogeneous & linear).

Let $k \in \mathbb{Z}^+$ & $c_n (\neq 0), c_{n-1}, c_{n-2} \ldots c_{n-k} (\neq 0)$ be real numbers. If $a_n, n \geq 0$ is a discrete function then

$$c_n a_n + c_{n-1} a_{n-1} + c_{n-2} a_{n-2} \ldots + c_{n-k} a_{n-k} = f(n), \quad n \geq k$$

$\Downarrow$

recurrence relation with constant coefficients of order k

Interest ourselves:

$$c_n a_n + c_{n-1} a_{n-1} + c_{n-2} a_{n-2} = 0, \quad n \geq 2.$$

Second order rr.

Trial soln:

Take one form first order. $\boxed{a_n = cr^n}$ where $c \neq 0, r \neq 0$.

$$c_n c r^n + c_{n-1} c r^{n-1} + c_{n-2} c r^{n-2} = 0$$

$$\boxed{c_n r^2 + c_{n-1} r + c_{n-2} = 0} \longrightarrow \text{Characteristic equation of a rr.}$$

Two roots r_1, r_2

1. r_1 & r_2 are distinct real numbers
2. r_1 & r_2 are real but $r_1 = r_2$
3. r_1 & r_2 are complex conjugate pairs.

(1)

$$a_n + a_{n-1} - 6a_{n-2} = 0 \qquad n \geq 2, \; a_0 = 1, \; a_1 = 2$$

$$\rightarrow r^2 + r - 6 = 0$$

$$(r+3)(r-2) = 0$$

$$r = -3, 2.$$

General soln: $\boxed{a_n = 2^n}$

$$\begin{array}{|l} c_1 2^n \\ c_2 (-3)^n \end{array}$$

General soln: $a_n = c_1 2^n + c_2 (-3)^n$.

$c_1 = ?$ $c_2 = ?$

$1 = a_0 = c_1 (2)^0 + c_2 (-3)^0 = c_1 + c_2$

$2 = a_1 = c_1 (2)^1 + c_2 (-3)^1 = 2c_1 - 3c_2$

$c_1 = 1, \; c_2 = 0$